

ANEESH GROVER

✉ agrover@rockets.utoledo.edu [in LinkedIn](#) [@ GitHub](#) [🌐 Portfolio](#)

Education

University of Toledo

Bachelor of Science in Computer Science and Engineering

Toledo, OH

GPA: 3.5/4.0

Relevant Coursework

- Data Structures and Algorithms
- Database Management Systems
- Operating Systems
- Computer Architecture
- Discrete Structures
- Artificial Intelligence

Experience

AcademicAlly (ACM Thapar Society Project) 🔗

July 2024 – August 2025

Software Developer

React, Python (Openpyxl, BeautifulSoup)

- Developed **AcademicAlly**, a responsive timetable portal serving **10,000+ students** for fast, reliable class schedule lookup.
- Converted **50,000+ Excel rows** into optimized JSON and engineered the backend using Python, Pandas, and Openpyxl, implementing **custom error handling** to reduce parsing errors by **95%**.
- Supported deployment and maintenance of the **live portal**, coordinating with team members.

Projects

Campus-Compute - Distributed Compute Framework 🔗

Python, FastAPI, WebSockets, Docker

- Architected a LAN-distributed compute system with a centralized coordinator and worker nodes, enabling automatic discovery and reconnect handling.
- Designed a capability-aware scheduler using CPU, RAM, and GPU metadata to optimize task placement across nodes.
- Built a real-time WebSocket dashboard for cluster telemetry, job orchestration, and containerized workload execution.

PatchIt - Civic Infrastructure Monitoring 🔗

PyTorch, YOLO, Depth-Anything-V2

- Co-led a crowdsourced platform with citizens and contractors to streamline reporting and resolution of road infrastructure issues.
- Built a YOLO-based pothole detection pipeline achieving **85.8% precision** and **81.7% mAP@50**, integrating depth estimation for 3D geometry and repair cost estimation.
- Runner-up (Open Innovation) at **NVIDIA SatHack Hackathon** among 1500+ participants.

DeepResearch - Multi-Agent Research Assistant 🔗

Streamlit, FAISS, HuggingFace, LLaMA3

- Built a multi-agent research system using LangGraph, integrating search APIs and LLMs for automated synthesis of structured, citation-backed insights.
- Implemented FAISS-based retrieval with embedding-driven indexing for context-aware query resolution.
- Designed a citation-aware RAG pipeline ensuring traceable outputs grounded in reliable sources.

Audio Deepfake Detection 🔗

PyTorch, AASIST-L

- Adapted and retrained a state-of-the-art audio deepfake detection model on a custom dataset, optimizing for mobile and edge deployment scenarios.
- Engineered training and inference pipelines by resolving bottlenecks, adding checkpointing, and improving stability under limited GPU resources.
- Achieved **3.37% Equal Error Rate** and **0.10 t-DCF**, demonstrating high accuracy in real-world spoof detection tasks.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Software Engineering & Systems: OOP, system design, REST APIs, asynchronous processing, version control

Machine Learning: PyTorch, TensorFlow, transformers, vector databases, RAG

Infrastructure & Tools: Docker, Git, CI/CD, Supabase, OpenCV, CUDA

Leadership & Involvement

Association for Computing Machinery (ACM) Thapar

Nov 2023 – Aug 2025

Executive Board (Community Collaboration) & Machine Learning Bootcamp Lead

- Organized ACM Thapar's flagship hackathon Hacklipse 2024, hosting 300+ participants.
- Mentored 100+ students through ACM's ML Bootcamp, designing curriculum and weekly sessions.

Achievements

- Secured **runner-up** in **Open Innovation Category** at **NVIDIA's SatHack**, competing against **1500+ participants**.
- Awarded Best Use Case at ACM Projectathon 2024 for AcademicAlly